RISC-V Cheatsheet

Arithmetic

Name	Syntax	Result	IF
add	add rd, rs1, rs2	rd = rs1 + rs2	R
addi	addi rd, rs1, imm	rd = rs1 + imm	I
sub	sub rd, rs1, rs2	rd = rs1 - rs2	R
lui	lui rd, imm	rd = imm $\ll$ 12	I
neg	neg rd, rs1	rd = -rs1	(p)
li	li rd, imm	rd = imm	(p)
mv	mv rd, rs1	rd = rs1	(p)
nop	nop	—	(p)

Logical

Name	Syntax	Result	IF
and	and rd, rs1, rs2	rd = rs1 & rs2	R
andi	andi rd, rs1, imm	rd = rs1 & imm	I
or	or rd, rs1, rs2	rd = rs1 | rs2	R
ori	ori rd, rs1, imm	rd = rs1 | imm	I
xor	xor rd, rs1, rs2	rd = rs1 $\oplus$ rs2	R
xori	xori rd, rs1, imm	rd = rs1 $\oplus$ imm	I
not	not rd, rs1	rd = $\sim$rs1	(p)

Shifts

Name	Syntax	Result	IF
sll	sll rd, rs1, rs2	rd = rs1 $\ll$ rs2	R
slli	slli rd, rs1, imm	rd = rs1 $\ll$ imm	I
srl	srl rd, rs1, rs2	rd = rs1 $\gg$ rs2	R
srli	srli rd, rs1, imm	rd = rs1 $\gg$ imm	I
sra	sra rd, rs1, rs2	rd = rs1 $\ggg$ rs2	R
srai	srai rd, rs1, imm	rd = rs1 $\ggg$ imm	I

Compares

Name	Syntax	Result	IF
slt	slt rd, rs1, rs2	rd = (rs1 < rs2)	R
slti	slti rd, rs1, imm	rd = (rs1 < imm)	I
sltu	sltu rd, rs1, rs2	rd = (rs1 < rs2)	R
sltiu	sltui rd, rs1, imm	rd = (rs1 < imm)	I
seqz	seqz rd, rs1	rd = (rs1 == 0)	(p)
snez	snez rd, rs1	rd = (rs1 $\neq$ 0)	(p)
sltz	sltz rd, rs1	rd = (rs < 0)	(p)
sgtz	sgtz rd, rs1	rd = (rs1 > 0)	(p)

Multiplication

Name	Syntax	Result	IF
mul	mul rd, rs1, rs2	rd = (rs1 * rs2)[31:0]	R
mulh	mulh rd, rs1, rs2	rd = (rs1 * rs2)[63:32]	R
mulhu	mulhu rd, rs1, rs2	rd = (rs1 * rs2)[63:32]	R
mulhsu	mulhsu rd, rs1, rs2	rd = (rs1 * rs2)[63:32]	R

Division

Name	Syntax	Result	IF
div	div rd, rs1, rs2	rd = rs1 / rs2	R
divu	divu rd, rs1, rs2	rd = rs1 / rs2	(p)
rem	rem rd, rs1, rs2	rd = rs1 % rs2	R
remu	remu rd, rs1, rs2	rd = rs1 % rs2	(p)

(p)	pseudo-instruction	rs1	source register 1
imm	immediate value	rs2	source register 2
pc	program counter	rd	destination register

Branches

Name	Syntax	Result	IF
beq	beq rs1, rs2, imm	if(rs1 == rs2): pc += imm	SB
bne	bne rs1, rs2, imm	if(rs1 $\neq$ rs2): pc += imm	SB
blt	blt rs1, rs2, imm	if(rs1 < rs2): pc += imm	SB
bltu	bltu rs1, rs2, imm	if(rs1 < rs2): pc += imm	SB
bge	bge rs1, rs2, imm	if(rs1 $\geq$ rs2): pc += imm	SB
bgeu	bgeu rs1, rs2, imm	if(rs1 $\geq$ rs2): pc += imm	SB
auipc	auipc rd, imm	rd = pc + (imm $\ll$ 12)	U
beqz	beqz rs1, imm	if(rs1 == 0): pc += imm	(p)
bnez	bnez rs1, imm	if(rs1 $\neq$ 0): pc += imm	(p)
bltz	bltz rs1, imm	if(rs1 < 0): pc += imm	(p)
bgt	bgt rs1, rs2, imm	if(rs1 > rs2): pc += imm	(p)
bgtu	bgtu rs1, rs2, imm	if(rs1 > rs2): pc += imm	(p)
bgtz	bgtz rs1, imm	if(rs1 > 0): pc += imm	(p)
ble	ble rs1, rs2, imm	if(rs1 $\leq$ rs2): pc += imm	(p)
bleu	bleu rs1, rs2, imm	if(rs1 $\leq$ rs2): pc += imm	(p)
blez	blez rs1, imm	if(rs1 $\leq$ 0): pc += imm	(p)
bgez	bgez rs1, imm	if(rs1 $\geq$ 0): pc += imm	(p)

Jumps

Name	Syntax	Result	IF
j	j imm	pc += imm	(p)
jal	jal rd, imm	rd = pc+4; pc += imm	UJ
jalr	jalr rd, rs1, imm	rd = pc+4; pc = rs1+imm	I

Functions

Name	Syntax	Result	IF
call	call label	ra = pc + 4; pc = &label	(p)
ret	ret	pc = ra	(p)

Loads

Name	Syntax	Result	IF
lw	lw rd, imm(rs1)	rd = mem[rs1+imm]	I
lh	lh rd, imm(rs1)	rd = mem[rs1+imm][0:15]	I
lhu	lhu rd, imm(rs1)	rd = mem[rs1+imm][0:15]	I
lb	lb rd, imm(rs1)	rd = mem[rs1+imm][0:7]	I
lbu	lbu rd, imm(rs1)	rd = mem[rs1+imm][0:7]	I
la	la rd, symbol	rd = &symbol	I

Stores

Name	Syntax	Result	IF
sw	sw rs2, imm(rs1)	mem[rs1+imm] = rs2	S
sh	sh rs2, imm(rs1)	mem[rs1+imm][0:15] = rs2	S
sb	sb rs2, imm(rs1)	mem[rs1+imm][0:7] = rs2	S

Registers

Register	Alias	Description
x0	zero	hard-wired zero
x1	ra	return address
x2	sp	stack pointer
x3	gp	global pointer
x4	tp	thread pointer
x5-7	t0-2	temporaries
x8	s0/fp	saved register / frame pointer
x9	s1	saved register
x10-11	a0-1	function arguments / return values
x12-16	a2-11	function arguments
x18-27	s2-11	saved registers
x28-31	t3-6	temporaries

Type	31	30 29 28 27 26 25	24 23 22 21	20	19 18 17 16 15	14 13 12	11 10 9 8	7	6 5 4 3 2 1 0
R	func7		rs2		rs1	func3	rd		opcode
I	imm[11:0]				rs1	func3	rd		opcode
S	imm[11:5]		rs2		rs1	func3	imm[4:0]		opcode
SB	imm[12]	imm[10:5]	rs2		rs1	func3	imm[4:1]	imm[11]	opcode
U	imm[32:12]						rd		opcode
UJ	imm[20]	imm[10:1]		imm[11]	imm[19:12]		rd		opcode